SUNCHENG XIANG

Assistant Professor
School of Biomedical Engineering
(co-affiliated with) Institute of Medical Robotics
Shanghai Jiao Tong University (SJTU)
Email: **xiangsuncheng17@sjtu.edu.cn** • **Google Scholar**
Personal Webpage: **https://JeremyXSC.github.io/**

Research Interests

Machine Learning and Computer Vision
Domain Adaptation, Image Generation
Medical Image Analysis, Robotic Surgery, 3D Reconstruction
Image Retrieval, Multimodal Foundation Models„ Representation Learning

Position

Shanghai Jiao Tong University, Shanghai, China
Assistant Professor — Aug 2022 - Present
School of Biomedical Engineering

Education

Shanghai Jiao Tong University, Shanghai, China — 2017 - 2022
Ph.D in Computer Science and Technology

National University of Defense Technology, Changsha, China — 2014 - 2017
M.S in Software Engineering

Changsha University of Science & Technology, Changsha, China — 2010 - 2014
B.S in Electrical Engineering and Automation

Preprints

[1]. SRPR-Net: Semantic and Relational Prompt Refinement for Automated SAM-based Instance Segmentation
arXiv preprint arXiv:2609.24226, 2026.
Lufei Liu, Guojie Li, **Suncheng Xiang**✉, Fan Zhang✉

[2]. CoBa: Cost-Effective Test-Time Scaling via Compute-Balanced Routing
arXiv preprint arXiv:2608.07424, 2026.
Yan Zhou, Yue Ouyang, Kaiyang Zheng, **Suncheng Xiang**✉

[3]. TEPA: Revoking Stale Memories for Conflict-Robust Language Agents
arXiv preprint arXiv:2608.07429, 2026.
Yan Zhou, Yue Ouyang, Kaiyang Zheng, **Suncheng Xiang**✉

[4]. ST-ColoNet: Spatio-Temporal Colon Segment Recognition via Hybrid Attention and Edge-Guided Feature Learning

arXiv preprint arXiv:2605.28119, 2026.
Ziyi Wang, Zhengjie Zhang, Jingsheng Gao, Dahong Qian, **Suncheng Xiang**✉

[5]. MedCore: Boundary-Preserving Medical Core Pruning for MedSAM
arXiv preprint arXiv:2605.13688, 2026.
Cenwei Zhang, **Suncheng Xiang**✉, Lei You

[6]. BMDS-Net: A Bayesian Multi-Modal Deep Supervision Network for Robust Brain Tumor Segmentation
arXiv preprint arXiv:2601.17504, 2026.
Yan Zhou†, Huang Zhen, Yingqiu Li, Ouyang Yue, **Suncheng Xiang†**, Zehua Wang

[7]. GPF-Net: Gated Progressive Fusion Learning for Polyp Re-Identification
arXiv preprint arXiv:2512.21476, 2025.
Suncheng Xiang✉, Xiaoyang Wang, Junjie Jiang, Hejia Wang, Dahong Qian

[8]. LDP: Parameter-Efficient Fine-Tuning of Multimodal LLM for Medical Report Generation
arXiv preprint arXiv:2512.10750, 2025.
Tianyu Zhou, Junyi Tang, Zehui Li, Dahong Qian, **Suncheng Xiang**✉

[9]. BALR-SAM: Boundary-Aware Low-Rank Adaptation of SAM for Resource-Efficient Medical Image Segmentation
arXiv preprint arXiv:2509.24204, 2025.
Zelin Liu, Sicheng Dong, Bocheng Li, Yixuan Yang, Jiacheng Ruan, Chenxu Zhou, **Suncheng Xiang**✉

[10]. Leveraging MIMIC Datasets for Better Digital Health: A Review on Open Problems, Progress Highlights, and Future Promises
arXiv preprint arXiv:2506.12808, 2025.
Afifa Khaled, Mohammed Sabir, Rizwan Qureshi, Camillo Maria Caruso, Valerio Guarrasi, **Suncheng Xiang**, S. Kevin Zhou

[11]. Generalizing Monocular Colonoscopy Image Depth Estimation by Uncertainty-based Global and Local Fusion Network
arXiv preprint arXiv:2409.15006, 2024.
Sijia Du, Chengfeng Zhou, **Suncheng Xiang**, Jianwei Xu, Dahong Qian

[12]. Learning Collaborative Knowledge with Multimodal Representation for Polyp Re-Identification
arXiv preprint arXiv:2408.05914, 2024.
Suncheng Xiang, Jiale Guan, Shilun Cai, Jiacheng Ruan, Dahong Qian

[13]. Towards Open-set Gesture Recognition via Feature Activation Enhancement and Orthogonal Prototype Learning
arXiv preprint arXiv:2308.00929, 2023.
Chen Liu, Can Han, Chengfeng Zhou, Crystal Cai, **Suncheng Xiang**, Hualiang Ni, Dahong Qian

[14]. Colo-ReID: Discriminative Representation Embedding with Meta-learning for Colonoscopic Polyp Re-Identification
arXiv preprint arXiv:2308.00929, 2023.
Suncheng Xiang✉, Chengfeng Zhou, Zhengjie Zhang, Shilun Cai, Dahong Qian

[15]. Attribute Analysis with Synthetic Dataset for Person Re-Identification
arXiv preprint arXiv:2006.07139, 2020.
Suncheng Xiang, Yuzhuo Fu, Guanjie You, Ting Liu

Journal Publications

[16]. PhysSFI-Net: Physics-informed Geometric Learning of Skeletal and Facial Interactions for Orthognathic Surgical Outcome Prediction
npj Digital Medicine (**npj Digit. Med.**), 2026. (**IF=18.0**)
Jiahao Bao, Huazhen Liu, et al., **Suncheng Xiang**✉, Hongbo Yu

[17]. The Current Status and Prospects of Artificial Intelligence in Digital Mental Health Research
Journal of Tongji University, (**J. Tongji Univ.**), 2026
Zijin Yu, Huiqi Li, Langyan Liu, **Suncheng Xiang**, Chunbo Li

[18]. ExFusion: Efficient Transformer Training via Multi-Experts Fusion
IEEE Transactions on Multimedia (**IEEE TMM**), 2026. (**IF=9.7**)
Jiacheng Ruan, Daize Dong, Xiaoye Qu, Tong Zhu, Ting Liu, Yuzhuo Fu, Yu Cheng, **Suncheng Xiang**✉

[19]. Ortho-OPD: an Automatic Osteotomy Planes Design Model for Orthognathic Surgery Based on Deep Learning
IEEE Journal of Biomedical and Health Informatics (**IEEE JBHI**), 2025. (**IF=6.7**)
Yiming Wang, Xiangmin Li, Yang Yang, Mengjia Cheng, Xu Zhang, Jiahao Bao, Hongjun Qian, Xinyi Huang, Zhihan Wu, **Suncheng Xiang**✉, Yaofeng Wen, Hongbo Yu

[20]. DiverSeed: Integrating Active Learning for Target Domain Data Generation in Instruction Tuning
Machine Learning (**ML**), 2025. (**IF=7.5**)
Jingsheng Gao, Mengnan Qi, Xian Gao, **Suncheng Xiang**, Ke Ji, Jiacheng Ruan, Ting Liu, Yuzhuo Fu

[21]. VM-UNet: Vision Mamba UNet for Medical Image Segmentation
ACM Transactions on Multimedia Computing, Communications and Applications (**ACM TOMM**), 2025. (**Highly Cited Paper**, **IF=5.1**)
Jiacheng Ruan, Jincheng Li, **Suncheng Xiang**✉

[22]. Learning Visual-Semantic Embedding for Generalizable Person Re-identification: A Unified Perspective
ACM Transactions on Multimedia Computing, Communications and Applications (**ACM TOMM**), 2025. (**IF=5.1**)
Suncheng Xiang✉, Jingsheng Gao, Mingye Xie, Mengyuan Guan, Jiacheng Ruan, Yuzhuo Fu

[23]. Learning Multi-axis Representation in Frequency Domain for Medical Image Segmentation
Machine Learning (**ML**), 2024. (**IF=7.5**)
Jiacheng Ruan, Jingsheng Gao, Mingye Xie, **Suncheng Xiang**✉

[24]. Deep Learning–Based Facial and Skeletal Transformations for Surgical Planning
Journal of Dental Research (**JDR**), 2024. (**IF=7.6**)
Jiahao Bao, Xiangning Zhang, **Suncheng Xiang**, et al.

[25]. Multi-step Validation of A Deep Learning-based System with Visual Explanations for Optical Diagnosis of Polyps with Advanced Features
iScience (**IS**), 2024. (**IF=4.1**)
Qing-Wei Zhang, Zhengjie Zhang, et al., **Suncheng Xiang**, et al.

[26]. SubFace: Learning with Softmax Approximation for Face Recognition
Multimedia Tools and Applications (**MTA**), 2024. (**IF=3.6**)
Suncheng Xiang✉, Hongwei Xu, Dahong Qian

[27]. Toward an End-to-End Implicit Addressee Modeling for Dialogue Disentanglement
Multimedia Tools and Applications (**MTA**), 2024. (**IF=3.6**)
Jingsheng Gao, Zeyu Li, **Suncheng Xiang**, Zhuowei Wang, Ting Liu, Yuzhuo Fu

[28]. A Simple Normalization Technique Using Window Statistics to Improve the Out-of-Distribution Generalization on Medical Images
IEEE Transactions on Medical Imaging (**IEEE TMI**), 2024. (**IF=10.6**)
Chengfeng Zhou, Jun Wang, **Suncheng Xiang**, Feng Liu, Hefeng Huang, Dahong Qian

[29]. Rethinking Person Re-Identification via Semantic-Based Pretraining
ACM Transactions on Multimedia Computing, Communications and Applications (**ACM TOMM**), 2023. (**IF=5.1**)
Suncheng Xiang✉, Dahong Qian, Jingsheng Gao, Zirui Zhang, Ting Liu, Yuzhuo Fu

[30]. Editing Outdoor Scenes with A Large Annotated Synthetic Dataset
Multimedia Tools and Applications (**MTA**), 2023. (**IF=3.6**)
Mingye Xie, Zongwei Liu, **Suncheng Xiang**, Ting Liu, Yuzhuo Fu

[31]. Deep Multimodal Representation Learning for Generalizable Person Re-identification
Machine Learning (**ML**), 2023. (**IF=7.5**)
Suncheng Xiang✉, Hao Chen, Wei Ran, Zefang Yu, Ting Liu, Dahong Qian, Yuzhuo Fu

[32]. Deep Learning-based PET/MR Radiomics for the Classification of Annualized Relapse Rate in Multiple Sclerosis
Multiple Sclerosis and Related Disorders (**MULT SCLER RELAT DIS**), 2023. (**IF=4.808**)
Sijia Du, et al., **Suncheng Xiang**, Dahong Qian, Biao Li, Sheng Chen, Min Zhang

[33]. Less is More: Learning from Synthetic Data with Fine-grained Attributes for Person Re-Identification
ACM Transactions on Multimedia Computing, Communications and Applications (**ACM TOMM**), 2023. (**IF=5.1**)
Suncheng Xiang✉, Dahong Qian, Mengyuan Guan, Binjie Yan, Ting Liu, Yuzhuo Fu, Guanjie You

[34]. Learning from Self-Discrepancy via Multiple Co-teaching for Cross-Domain Person Re-Identification
Machine Learning (**ML**), 2022. (**Invited Paper, IF=7.5**)
Suncheng Xiang✉, Yuzhuo Fu, Mengyuan Guan, Ting Liu

[35]. Multi-level Feature Learning with Attention for Person Re-Identification
Multimedia Tools and Applications (**MTA**), 2020. (**IF=3.6**)
Suncheng Xiang, Yuzhuo Fu, Hao Chen, Wei Ran, Ting Liu

[36]. Progressive Learning with Style Transfer for Distant Domain Adaptation
IET Image Processing (**IET-IPR**), 2020. (**IF=2.373**)
Suncheng Xiang, Yuzhuo Fu, Ting Liu

[37]. Unsupervised Person Re-Identification by Hierarchical Cluster and Domain Transfer
Multimedia Tools and Applications (**MTA**), 2020. (**IF=3.6**)
Suncheng Xiang, Yuzhuo Fu, Mingye Xie, Zefang Yu, Ting Liu

Conference Publications

[38]. Trusted Multi-View Deep Learning Classification of Fetal Congenital Heart Disease with Feature-level and Decision-level Fusion
International Symposium on AI and Cybersecurity (**ISAICS**), 2026.
Tan Zhou, Shifa Yao, **Suncheng Xiang**, Dahong Qian, Baoying Ye

[39]. MicroAUNet: Boundary-Enhanced Multi-scale Fusion with Knowledge Distillation for Colonoscopy Polyp Image Segmentation
European Conference on Computer Vision (**MedVidU @ ECCV**), 2026.
Ziyi Wang, Yuanmei Zhang, Baoying Ye, Yimei Jiang, Leilei Gu, **Suncheng Xiang**✉

[40]. Visual Encoders for Generalized Chromosome Recognition
International Conference on Image Processing (**ICIP**), 2025.
Ruijia Chang, Tao Zhou, **Suncheng Xiang**, Yujia Wang, Kui Su, Yin Zhou, Dahong Qian, Jun Wang

[41]. Efficient Gas Source Tracking in Turbulent Environments Using a Probabilistic Vision Transformer
International Conference on Electronics, Communications and Intelligent Science (**ECIS**), 2025.
Zhituo Geng, Yu Fu, **Suncheng Xiang**, Zehui Li

[42]. Bridging the Gap: Balancing Human Perception and Detector Attention in Adversarial Attacks
IEEE International Conference on Multimedia and Expo (**ICME**), 2025.
Mingye Xie, **Suncheng Xiang**, Xian Gao, Ting Liu, Yuzhuo Fu

[43]. A Training-Free Correlation-Weighted Model for Zero-/Few-Shot Industrial Anomaly Detection with Retrieval Augmentation
IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**), 2025.
Wei Ran, Zefang Yu, **Suncheng Xiang**, Ting Liu, Yuzhuo Fu

[44]. GPA: Enhancing Generalizable Physical Adversarial Attacks Across Multiple Vision Tasks
IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**), 2025.
Mingye Xie, **Suncheng Xiang**, Jiacheng Ruan, Xian Gao, Zefang Yu, Ting Liu, Yuzhuo Fu

[45]. TTE: Two Tokens are Enough to Improve Parameter-Efficient Tuning
Thirty-Ninth AAAI Conference on Artificial Intelligence (**AAAI**), 2025. (**Accept rate~23.4%**)
Jiacheng Ruan, Mingye Xie, Jingsheng Gao, Xian Gao, **Suncheng Xiang**, Ting Liu, Yuzhuo Fu

[46]. GIST: Improving Parameter Efficient Fine-Tuning via Knowledge Interaction
ACM International Conference on Multimedia (**ACM MM**), 2024. (**Accept rate~26.2%**)
Jiacheng Ruan, Jingsheng Gao, Mingye Xie, **Suncheng Xiang**, Zefang Yu, Ting Liu, Yuzhuo Fu, Xiaoye Qu

[47]. Oceanship: A Large-Scale Dataset for Underwater Audio Target Recognition
International Conference On Intelligent Computing (**ICIC**), 2024. (**Oral**)
Zeyu Li, **Suncheng Xiang**, Tong Yu, Jingsheng Gao, Jiacheng Ruan, Yanping Hu, Ting Liu, Yuzhuo Fu

[48]. iDAT: inverse Distillation Adapter-Tuning
IEEE International Conference on Multimedia and Expo (**ICME**), 2024. (**Oral**)
Jiacheng Ruan, Jingsheng Gao, Mingye Xie, Daize Dong, **Suncheng Xiang**, Ting Liu, Yuzhuo Fu

[49]. Enhancing Nasopharyngeal Carcinoma Classification Based on Multi-View Cross-Modal Knowledge Distillation
IEEE International Symposium on Biomedical Imaging (**ISBI**), 2024.
Zhengjie Zhang, Crystal Cai, Sijia Du, **Suncheng Xiang**, Dahong Qian

[50]. VT-ReID: Learning Discriminative Visual-Text Representation for Polyp Re-Identification
IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**), 2024.
(**Accept rate~45%**, **Oral**)
Suncheng Xiang✉, Cang Liu, Jiacheng Ruan, Shilun Cai, Sijia Du, Dahong Qian

[51]. LAMM: Label Alignment for Multi-Modal Prompt Learning
Thirty-Eighth AAAI Conference on Artificial Intelligence (**AAAI**), 2024. (**Accept rate~23.75%**)
Jingsheng Gao, Jiacheng Ruan, **Suncheng Xiang**, Zefang Yu, Ke Ji, Mingye Xie, Ting Liu, Yuzhuo Fu

[52]. SAE-NTM: Sentence-Aware Encoder for Neural Topic Modeling
4th Workshop on Computational Approaches to Discourse (**CODI**), 2023.
Hao Liu, Jingsheng Gao, **Suncheng Xiang**, Ting Liu, Yuzhuo Fu

[53]. CluCDD: Contrastive Dialogue Disentanglement via Clustering
IEEE International Conference on Acoustics, Speech and Signal Processing Satellite Workshop (**IWCIM**), 2023.
Jingsheng Gao, Zeyu Li, **Suncheng Xiang**, Ting Liu, Yuzhuo Fu

[54]. Colo-SCRL: Self-Supervised Contrastive Representation Learning for Colonoscopic Video Retrieval
IEEE International Conference on Multimedia and Expo (**ICME**), 2023. (**Oral**)
Qingzhong Chen, Shilun Cai, Crystal Cai, Zefang Yu, Dahong Qian✉, **Suncheng Xiang**✉

[55]. AutoKary2022: A Large-Scale Densely Annotated Dateset for Chromosome Instance Segmentation
IEEE International Conference on Multimedia and Expo (**ICME**), 2023.
Dan You, Pengcheng Xia, Qiuzhu Chen, Minghui Wu, **Suncheng Xiang**✉, Jun Wang✉

[56]. AV-TAD: Audio-Visual Temporal Action Detection with Transformer
IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**), 2023. (**Accept rate~40+%**, **Oral**)
Yangcheng Li, Zefang Yu, **Suncheng Xiang**, Ting Liu, Yuzhuo Fu

[57]. MTDL-Net: Morphological and Temporal Discriminative Learning for Heartbeat Classification
IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**), 2023. (**Accept rate~40+%**)
Can Han, **Suncheng Xiang**✉, Dahong Qian✉

[58]. CC-PoseNet: Towards Human Pose Estimation in Crowded Classrooms
IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**), 2023. (**Accept rate~40+%**, **Oral**)
Zefang Yu, Yanping Hu, **Suncheng Xiang**, Ting Liu, Yuzhuo Fu

[59]. MALUNet: A Muti-Attention and Light-weight UNet for Skin Lesion Segmentation
International Conference on Bioinformatics and Biomedicine (**BIBM**), 2022. (**Accept rate~19.8%**)
Jiacheng Ruan, **Suncheng Xiang**✉, Mingye Xie, Ting Liu, Yuzhuo Fu✉

[60]. CDTnet: Cross-Domain Transformer based on Attributes for Person Re-Identification
IEEE International Conference on Multimedia and Expo Workshops (**ICMEW**), 2022.
Mengyuan Guan, **Suncheng Xiang**, Ting Liu, Yuzhuo Fu

[61]. Rethinking Illumination for Person Re-Identification: A Unified View
IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (**CVPRW**), 2022.
Suncheng Xiang, Guanjie You, Leqi Li, Mengyuan Guan, Ting Liu, Dahong Qian, Yuzhuo Fu

[62]. Spatial Attention Guided Local Facial Attribute Editing
IEEE International Conference on Multimedia and Expo (**ICME**), 2022. (**Accept rate~29%**)
Mingye Xie, **Suncheng Xiang**, Feng Wang, Ting Liu, Yuzhuo Fu

[63]. Zero-Shot Learning for Skeleton-based Classroom Action Recognition
International Symposium on Computer Science and Intelligent Controls (**ISCSIC**), 2021.
Bin Shi, Luyang Wang, Zefang Yu, **Suncheng Xiang**, Yuzhuo Fu, Ting Liu

[64]. Learning from Self-Discrepancy via Multiple Co-teaching for Cross-Domain Person Re-Identification
International Joint Conference on Artificial Intelligence WSRL Workshop (**IJCAIW**), 2021.
Suncheng Xiang, Yuzhuo Fu, Mengyuan Guan, Ting Liu

[65]. Attention based Facial Expression Manipulation
IEEE International Conference on Multimedia and Expo Workshops (**ICMEW**), 2021.
Feng Wang, **Suncheng Xiang**, Ting Liu, Yuzhuo Fu

[66]. Taking a Closer Look at Synthesis: Fine-grained Attribute Analysis for Person Re-Identification
IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**), 2021. (**Accept rate~48%**)
Suncheng Xiang, Yuzhuo Fu, Guanjie You, Ting Liu

[67]. Unsupervised Domain Adaptation Through Synthesis for Person Re-Identification
IEEE International Conference on Multimedia and Expo (**ICME**), 2020. (**Accept rate~29%**)
Suncheng Xiang, Yuzhuo Fu, Guanjie You, Ting Liu

[68]. Deep Unsupervised Progressive Learning for Distant Domain Adaptation
IEEE International Conference on Tools with Artificial Intelligence (**ICTAI**), 2019. (**Accept rate~28%**, **Oral**)
Suncheng Xiang, Yuzhuo Fu, Ting Liu

[69]. Enhancing Model Performance of Person Re-Indentification on Unknown Target Domain
International Conference on Software Engineering and Service Science (**ICSESS**), 2018.
Rongsen Xu, Yuzhuo Fu, Ting Liu, **Suncheng Xiang**

[70]. Circuit design and optimization based on EXT, EXTU instruction
International Conference on Design, Manufacturing and Mechatronics (**ICDMM**), 2016.
Suncheng Xiang, Minxuan Zhang, Zuocheng Xing, Cang Liu

[71]. QRD Architecture Using the Modified ILMGS Algorithm for MIMO Systems
International Wireless Internet Conference (**WICON**), 2016.
Cang Liu, Chuan Tang, Zuocheng Xing, Luechao Yuan, Yu Wang, Lirui Chen, Yang Zhang, **Suncheng Xiang**, Wangfeng Zhao, Xing Hu, Jinsong Xu

[72]. Hardware design of ML algorithm in MIMO-OFDM system
International Conference on Systems and Informatics (**ICSAI**), 2016.
Suncheng Xiang, Minxuan Zhang, Zuocheng Xing, Cang Liu

Patents & Monographs

[73]. A Method, Equipment and Storage Medium for Re-Identification of Inland Water Vessels Based on Transfer Learning
National Invention Patent. Patent No: CN202010053647.6
Yuzhuo Fu, Ting Liu, **Suncheng Xiang**

[74]. Reusable Implantable Drive Mechanism Device
National Invention Patent. Patent No: CN202311730617.4
Ding Han, Dahong Qian, Guozheng Yan, Shuai Kuang, **Suncheng Xiang**, et al.

[75]. Artificial Intelligence Applications in Medicine
Shanghai Scientific and Technical Publishers (978-7-5478-7003-7)
Dahong Qian, **Suncheng Xiang**, et al.

Awards

- ESWA Excellent Reviewer, Elsevier, 2026
- IEEE TMI Distinguished Reviewer, IEEE, 2025
- Best Researcher Award, ScienceFather, 2025
- Third Prize Award in Young Teachers' Teaching Competition, SJTU, 2025
- Third Prize Award in Young Teachers' Teaching Competition, SJTU, 2024
- IEEE TMI Distinguished Reviewer, IEEE, 2024
- Meng Minwei International Exchange Fund, SJTU, 2024
- Outstanding Doctoral Graduate, SJTU, 2022
- Leo KoGuan Scholarship, SJTU, 2019 - 2020
- Merit Student, SJTU, 2018-2019
- First-class Academic Scholarship, NUDT, 2016-2017
- Outstanding Student, NUDT, 2015-2016
- Outstanding Student, NUDT, 2014-2015
- Third-class Academic Scholarship, Outstanding League Cadres, CSUST, 2012-2013
- First-class Academic Scholarship, Merit Student, Model Student of Academic Records, CSUST, 2011-2012
- Third-class Academic Scholarship, Excellent League Member, CSUST, 2010-2011

Projects

- Teaching Development Fund, 2025 - 2026
- Special Fund for "AI + Education" in SJTU, 2024 - 2025
- Scientific Research Project of Shanghai Municipal Health Commission, 2023 - 2026
- Youth Project of National Natural Science Foundation of China, 2024 - 2026
- Meng Minwei International Exchange Fund, 2024 - 2025
- Startup Fund for Young Faculty at SJTU, 2023 - 2025

Professional Activities

Journal Review

- IEEE Transactions on Image Processing
- IEEE Transactions on Medical Imaging (**Distinguished Reviewer**)
- IEEE Transactions on Intelligent Vehicles
- IEEE Transactions on Multimedia
- IEEE Transactions on Neural Networks and Learning Systems
- IEEE Transactions on Circuits and Systems for Video Technology
- IEEE Transactions on Information Forensics and Security
- IEEE Transactions on Biomedical Engineering
- CAAI Transactions on Intelligence Technology
- ACM Transactions on Multimedia Computing, Communications and Applications
- IEEE Journal of Biomedical and Health Informatics
- Information Processing and Management
- Pattern Recognition
- Machine Learning
- Machine Intelligence Research
- Information Fusion
- Defense Technology
- IET Image Processing
- International Journal of Intelligent Systems
- Signal Processing: Image Communication
- Expert Systems with Applications (**Excellent Reviewer**)
- Pattern Recognition Letters
- Computer Vision and Image Understanding
- Neurocomputing
- JMIR Medical Informatics
- Franklin Open
- Machine Vision and Applications

- Multimedia Systems
- Pattern Analysis and Applications
- Knowledge-Based Systems
- Multimedia Tools and Applications
- IEEE Access
- Artificial Intelligence In Medicine
- Biomedical Signal Processing and Control
- Applied Intelligence
- Journal of Imaging
- Data in Brief
- Journal of Electronic Imaging
- Journal of Supercomputing
- IEEE Open Journal of Signal Processing
- Computers, Materials and Continua
- Mathematical Biosciences and Engineering
- Computer Methods and Programs in Biomedicine
- Biomedicines
- Image and Vision Computing
- Results in Engineering
- Scientific Reports
- Discover Artificial Intelligence
- Journal of Imaging Informatics in Medicine

Journal Editorial

- Associate Editor, Scientific Reports, since 2026.
- Associate Editor, Medical Gas Research, since 2026.
- Associate Editor, Aging Advances, since 2026.
- Guest Editor, Advances in Differential Equations and Control Processes, since 2025.

- Associate Editor, Intelligent Manufacturing, since 2024.
- Associate Editor, Biomedical Engineering Communications, since 2024.
- Associate Editor, Electronics and Signal Processing, since 2024.
- Associate Editor, Open Access Journal of Data Science and Artificial Intelligence, since 2023.
- Associate Editor, International Journal of Architectural Engineering Technology, since 2023.

Conference Review

- CVPR 2024-2026
- ICCV 2025
- ECCV 2026
- BMVC 2026
- ACM MM 2023-2025
- NeurIPS 2025-2026
- AAAI 2026-2027
- ICDM 2026
- MICCAI 2024
- BIBM 2026
- CODI-CRAC 2026
- ISBI 2024-2026
- IJCNN 2025
- WCCI 2026
- ICRA 2025
- ICME 2023-2026
- ICASSP 2023-2027
- CECNet 2022
- CONF-MPCS 2026
- IWACCE 2022
- IEEE EMBS BHI 2026

Program Committee

- Area Chair of NeurIPS 2025
- Session Chair of ACML 2024
- Session Chair of ICASSP 2024
- CTMDT 2024-2028
- EACL 2024
- Workshop Chair of CVDL 2024
- ACL 2023
- Session Chair of ICME 2022 (Recognition and Retrieval)
- EMNLP 2022-2025
- ML4CS 2022
- Session Chair of ICTAI 2019

Professional Membership

- Member, CSIG, (2026-)
- Member, CCF-CV, (2025-)
- Member, IEEE, (2022-)
- Member, ACM, (2023-)
- Student Member, CCF (2018-2021)

Skills

* Programming: Python, MATLAB, C/C++
* Deep Learning: Pytorch, TensorFlow, Caffe

Teaching

- Medical AI applications, SJTU (Spring 2025-2026)
- Artificial Intelligence Foundation A, SJTU (Spring 2025-2026)
- Artificial Intelligence and Medicine, SJTU (Spring 2024-2025)
- Medical Imaging Intelligence Technology, SJTU (Spring 2024-2026)

- Artificial Intelligence Interaction Techniques, SJTU (Spring 2023-2024)
- Artificial Intelligence and Medicine, SJTU (Spring 2023-2024)
- Artificial Intelligence and Medicine, SJTU (Spring 2022-2023)
- BioDesign, SJTU (Fall 2022-2023)
- Digital Integrated Circuits, SJTU (Lead TA, Fall 2020-2021)
- Operating System, SJTU (Lead TA, Fall 2018-2019)
- Digital Integrated Circuits, SJTU (Lead TA, Spring 2017-2018)
- Fundamentals of College Computer, NUDT (Lead TA, Fall 2015-2016)

Invited Talks

- 2026.10: Generative AI and Multimodal Learning: From Foundation Models to Embodied Intelligence (XJTLU)
- 2026.06: Multimodal Medical Processing and Navigation, China Society of Image and Graphics (CSIG)
- 2025.05: Generative Artificial Intelligence and Multimodal Learning: Ushering in a New Era of Innovation, Shanghai Jiao Tong University
- 2024.06: Multimodal Cognitive Learning Based on Generative Artificial Intelligence, Shanghai Theatre Academy
- 2023.11: The 6th WLA Forum, Shanghai
- 2023.08: Summer Top Science Exploration Camp for ”Future Scientists” Training Program, Shanghai
- 2023.03: T-Workshop of Science and Technology for the Common Destiny of Mankind, Shanghai
- 2022.04: Pedestrian Data Synthesis and Application Based on Deep Learning, Wuhan University
- 2022.03: Learning from Synthetic Data with Fine-grained Attributes, Fudan University
- 2021.12: Leveraging Synthetic Data for Person Re-Identification, Hunan University
- 2021.12: Learning from Synthetic Data for Person Re-Identification, Donghua University

Open Source

Codes and models for my published papers are available on my GitHub:
https://github.com/JeremyXSC/

Last updated: October 27, 2026